StudySmart AI: Predicting Secondary School Final Grades

Short reproducible research report | 18 October 2026

Research question

How accurately can final grades be predicted from school/lifestyle context, and how much does adding previous grades improve performance?

Data and target

We analyzed 649 Portuguese-language course records from the UCI Student Performance dataset, drawn from two Portuguese secondary schools. The target G3 is the final grade on a 0–20 scale. Inputs include study-time category, absences, free time, social activity, support and health. The history model also uses G1 and G2 (earlier period grades). Sleep hours and screen time are not present in this dataset.

Methods

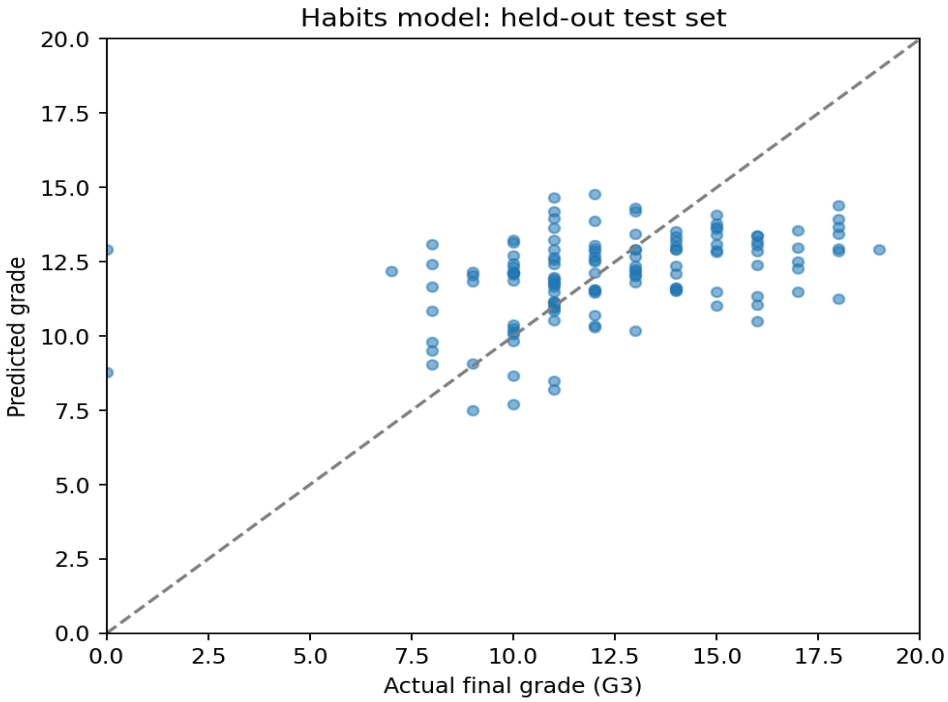
We used a reproducible random 80/20 split (seed 42): 519 training and 130 held-out test records. Within the training split, five-fold cross-validation selected the lowest-MAE model from a mean baseline, Ridge regression, Random Forest and Histogram Gradient Boosting. Preprocessing was performed inside each cross-validation pipeline. Predictions were clipped to the observed grade range (0–20) for test evaluation. A single test set was used, without external validation.

Held-out results

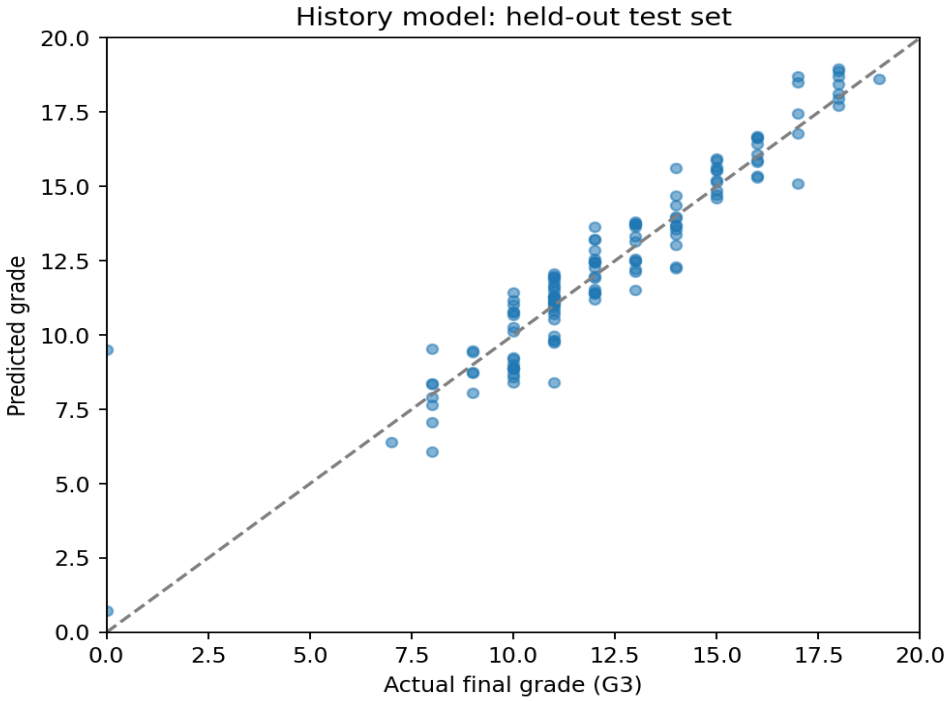
Feature set	Model	CV MAE	Test MAE	RMSE	R^2
Habits/context	ridge	2.154	2.157	2.873	0.153
+ previous grades	ridge	0.831	0.754	1.182	0.857

Adding previous grades reduced held-out MAE from 2.16 to 0.75 grade points, a 65.1% relative reduction. Both models selected Ridge regression by cross-validation. This is predictive evidence, not proof that any measured behavior causes a change in grades.

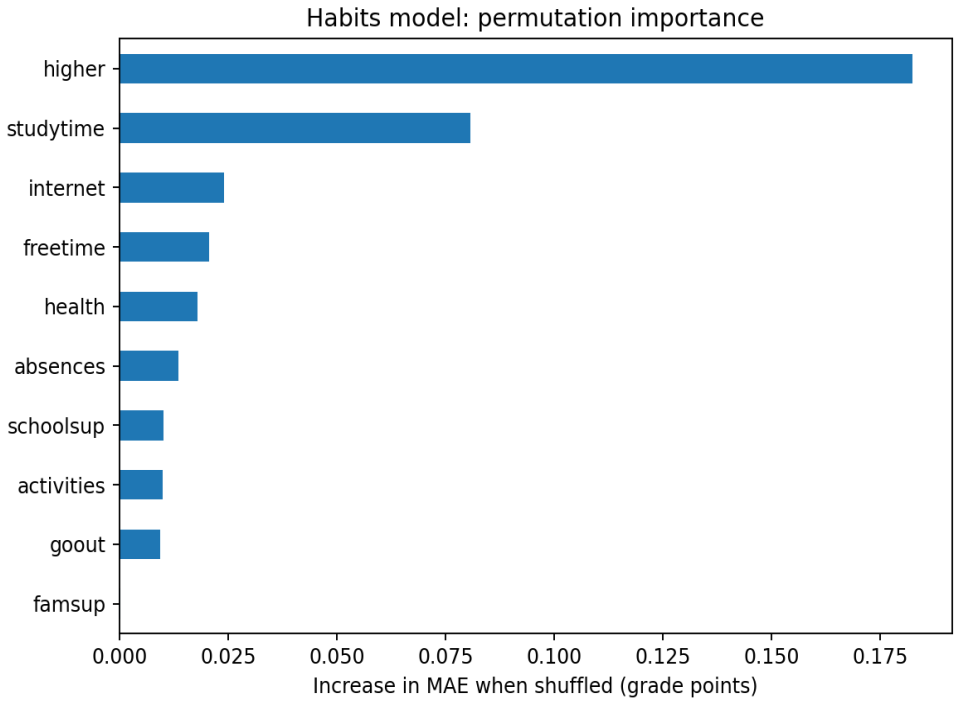
Habits/context: predictions



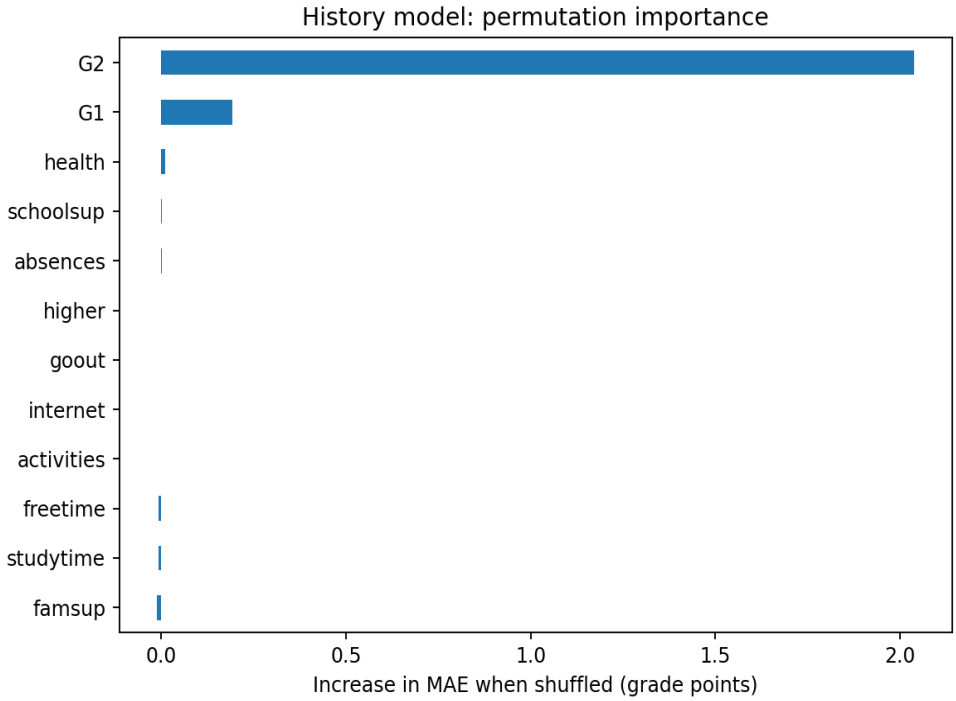
With previous grades: predictions



Habits/context: permutation importance



With previous grades: permutation importance



Limitations and responsible use

The sample is small, historical, from two schools in one country, and observational. The habits/context model explains relatively little test variance. Prior grades are only available later in the academic year. Some inputs may encode socioeconomic context. Permutation importance indicates reliance by this fitted model, not causal influence. We did not perform external validation, fairness auditing or uncertainty calibration. The system should not be used for student selection, discipline or individualized high-stakes academic decisions.

Reproducibility and source

All experiment scripts, dataset documentation, metrics, figures and app code are packaged with this report. Re-run `python train.py --data data/student-por.csv` to reproduce the fixed-seed experiment. Source: Cortez, P. and Silva, A. (2008), Using Data Mining to Predict Secondary School Student Performance; UCI Machine Learning Repository, Student Performance, https://archive.ics.uci.edu/dataset/320/student+performance. The project was developed with AI assistance; authorship claims should reflect actual contributions.